

cases a smaller, suboptimal filter (called a reduced-order filter) can be designed to give acceptable estimation performance at a lower computational cost. Section 10.4 discusses robust Kalman filtering, which is a way of making the filter less sensitive to variations in the assumed system model. Section 10.5 discusses the topic of delayed measurements. Sometimes the measurements do not arrive at the filter in chronological order because of processing delays. In these cases, we can modify the filter to optimally incorporate measurements that arrive at the filter in the wrong sequence.

10.1 VERIFYING KALMAN FILTER PERFORMANCE

We can verify Kalman filter performance, or adjust the gain of the Kalman filter, using our knowledge of the statistics of the innovations. The innovations is defined as $(y_k - H_k \hat{x}_k^-)$, and in this section we will show that it is a zero-mean white stochastic process with a covariance of $(H_k P_k^- H_k^T + R_k)$.

Recall our original system model, along with the one-step *a priori* update equation for the state estimate:

$$\begin{aligned} x_k &= F_{k-1} x_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ \hat{x}_{k+1}^- &= F_k \hat{x}_k^- + F_k K_k (y_k - H_k \hat{x}_k^-) \end{aligned} \quad (10.1)$$

The innovations is defined as the quantity in parentheses in the update equation. The innovations can be thought of as the part of the measurement that contains new information and that is therefore used to update the state estimate (apart from our knowledge of the state transition matrix). If the innovations was zero then the state estimate would simply be updated according to the state transition matrix. A nonzero innovations allows the measurement to affect the state estimate. The innovations r_k can be written as

$$\begin{aligned} r_k &= y_k - H_k \hat{x}_k^- \\ &= (H_k x_k + v_k) - H_k \hat{x}_k^- \\ &= H_k (x_k - \hat{x}_k^-) + v_k \\ &= H_k \epsilon_k + v_k \end{aligned} \quad (10.2)$$

where ϵ_k , the *a priori* estimation error, is defined by the above equation. The covariance of the innovations is given as

$$E[r_k r_i^T] = E[(H_k \epsilon_k + v_k)(H_i \epsilon_i + v_i)^T] \quad (10.3)$$

Let us see what the covariance is when $k \neq i$. We can assume without loss of generality that $k > i$. We then obtain

$$E[r_k r_i^T] = H_k E(\epsilon_k \epsilon_i^T) H_i^T + H_k E(\epsilon_k v_i^T) \quad (10.4)$$

Note that two of the cross terms reduced to zero because of the whiteness of v_k , and the fact that the estimation error ϵ_i is independent of v_k for $k > i$. In order to evaluate this covariance, we need to evaluate $E(\epsilon_k \epsilon_i^T)$ and $E(\epsilon_k v_i^T)$. First we

will evaluate $E(\epsilon_k \epsilon_i^T)$. In order to evaluate this term, notice that the *a priori* state estimate can be written as follows:

$$\begin{aligned}\hat{x}_{k+1}^- &= F_k \hat{x}_k^- + F_k K_k (y_k - H_k \hat{x}_k^-) \\ &= F_k \hat{x}_k^- + F_k K_k (H_k x_k + v_k - H_k \hat{x}_k^-) \\ &= F_k \hat{x}_k^- + F_k K_k H_k (x_k - \hat{x}_k^-) + F_k K_k v_k\end{aligned}\tag{10.5}$$

The *a priori* estimation error can be written as

$$\begin{aligned}\epsilon_{k+1} &= x_{k+1} - \hat{x}_{k+1}^- \\ &= F_k (x_k - \hat{x}_k^-) - F_k K_k H_k (x_k - \hat{x}_k^-) + w_k - F_k K_k v_k \\ &= F_k (I - K_k H_k) \epsilon_k + (w_k - F_k K_k v_k) \\ &= \tilde{\phi}_k \epsilon_k + v'_k\end{aligned}\tag{10.6}$$

where $\tilde{\phi}_k$ and v'_k are defined by the above equation. This is a linear discrete-time system for ϵ_k with the state transition matrix

$$\tilde{\phi}_{k,i} = \begin{cases} \tilde{\phi}_{k-1} \tilde{\phi}_{k-2} \cdots \tilde{\phi}_i & k > i \\ I & k = i \end{cases}\tag{10.7}$$

ϵ_k can be solved from the initial condition ϵ_i as follows:

$$\epsilon_k = \tilde{\phi}_{k,i} \epsilon_i + \sum_{j=i}^{k-1} \tilde{\phi}_{k,j+1} v'_j\tag{10.8}$$

The covariance of $\epsilon_k \epsilon_i^T$ can be written as

$$E(\epsilon_k \epsilon_i^T) = E \left[\left(\tilde{\phi}_{k,i} \epsilon_i + \sum_{j=i}^{k-1} \tilde{\phi}_{k,j+1} v'_j \right) \epsilon_i^T \right]\tag{10.9}$$

We see that all of the $v'_j \epsilon_i^T$ terms in the above expression are zero-mean. This is because all of the v'_j noise terms occur at time i or later and so do not affect ϵ_i . [Note from Equation (10.6) that ϵ_i is affected only by the noise terms at time $(i-1)$ or earlier.] Therefore,

$$E(v'_j \epsilon_i^T) = 0 \quad (j \geq i)\tag{10.10}$$

We therefore see that Equation (10.9) can be written as

$$\begin{aligned}E(\epsilon_k \epsilon_i^T) &= \tilde{\phi}_{k,i} E(\epsilon_i \epsilon_i^T) \\ &= \tilde{\phi}_{k,i} P_i^-\end{aligned}\tag{10.11}$$

Now that we have computed $E(\epsilon_k \epsilon_i^T)$, we need to solve for $E(\epsilon_k v_i^T)$ in order to arrive at our goal, which is the evaluation of Equation (10.4). $E(\epsilon_k v_i^T)$ can be written as

$$E(\epsilon_k v_i^T) = E \left[\left(\tilde{\phi}_{k,i} \epsilon_i + \sum_{j=i}^{k-1} \tilde{\phi}_{k,j+1} v'_j \right) v_i^T \right]\tag{10.12}$$

The $\epsilon_i v_i^T$ term in the above expression is zero-mean, and the $v_j' v_i^T$ terms are zero-mean for $j > i$. The above covariance can therefore be written as

$$\begin{aligned} E(\epsilon_k v_i^T) &= E(\tilde{\phi}_{k,i+1} v_i^T) \\ &= E(\tilde{\phi}_{k,i+1} (w_i - F_i K_i v_i)^T) \\ &= -\tilde{\phi}_{k,i+1} F_i K_i R_i \end{aligned} \quad (10.13)$$

Substituting this equation, along with Equation (10.11), into Equation (10.4) gives

$$\begin{aligned} E(r_k r_i^T) &= H_k E(\epsilon_k \epsilon_i^T) H_i^T + H_k E(\epsilon_k v_i^T) \\ &= H_k \tilde{\phi}_{k,i} P_i^- H_i^T - H_k \tilde{\phi}_{k,i+1} F_i K_i R_i \\ &= H_k \tilde{\phi}_{k,i+1} (\tilde{\phi}_i P_i^- H_i^T - F_i K_i R_i) \end{aligned} \quad (10.14)$$

Now use the fact from Equation (10.6) that $\tilde{\phi}_i = F_i(I - K_i H_i)$ to obtain

$$\begin{aligned} E(r_k r_i^T) &= H_k \tilde{\phi}_{k,i+1} (F_i P_i^- H_i^T - F_i K_i H_i P_i^- H_i^T - F_i K_i R_i) \\ &= H_k \tilde{\phi}_{k,i+1} [F_i P_i^- H_i^T - F_i K_i (H_i P_i^- H_i^T + R_i)] \end{aligned} \quad (10.15)$$

Now use the fact that $K_i = P_i^- H_i^T (H_i P_i^- H_i^T + R_i)^{-1}$ (the standard Kalman filter gain equation) to obtain

$$\begin{aligned} E(r_k r_i^T) &= H_k \tilde{\phi}_{k,i+1} (F_i P_i^- H_i^T - F_i P_i^- H_i^T) \\ &= 0 \quad \text{for } k > i \end{aligned} \quad (10.16)$$

So we see that the innovations r_k is white noise. Our next task is to determine its covariance. In order to do this we write the covariance as

$$\begin{aligned} E(r_k r_k^T) &= E[(y_k - H_k \hat{x}_k^-)(y_k - H_k \hat{x}_k^-)^T] \\ &= E\{[H_k(x_k - \hat{x}_k^-) + v_k][H_k(x_k - \hat{x}_k^-) + v_k]^T\} \\ &= H_k E(\epsilon_k \epsilon_k^T) H_k^T + E(v_k v_k^T) \\ &= H_k P_k^- H_k^T + R_k \end{aligned} \quad (10.17)$$

We therefore see that the innovations is a white noise process with zero mean and a covariance of $(H_k P_k^- H_k^T + R_k)$. While the Kalman filter is operating, we can process the innovations, compute its mean and covariance, and verify that it is white with the expected mean and covariance. If it is colored, nonzero-mean, or has the wrong covariance, then there is something wrong with the filter. The most likely reason for such a discrepancy is a modeling error. In particular, an incorrect value of F , H , Q , or R could cause the innovations to statistically deviate from its theoretically expected behavior. Statistical methods can then be used to tune F , H , Q , and R in order to force the innovations to be white zero-mean noise with a covariance of $(H_k P_k^- H_k^T + R_k)$ [Meh70, Meh72]. This concept is illustrated in Figure 10.1. A scalar example is presented in Problem 10.1.

Alternatively, if the engineer is uncertain of the correct values of F , H , Q , and R , then a bank of Kalman filters can be run in parallel, each Kalman filter with a value of F , H , Q , and R that the engineer thinks may be likely. Then the innovations can be inspected in each filter, and the one that matches theory is assumed to have